

Section J Annex 1503050 Spread Sheets will be attached as separate attachments to this solicitation:

1. J-1503050-08 Bethesda
2. J-1503050-08 Quantico
3. J-1503050-08 Washington
4. J-1503050-09(a) Irrigation System Inventory
5. J-1503050-11(a) Washington SWD System Inventory
6. J-1503050-11(f) Bethesda SWD System Inventory

J-1503050-13(a)

Storm Water Management Low Impact Development (LID) Maintenance (WNY)

BACKGROUND

In an effort to improve the quality and reduce the quantity and velocity of storm water runoff from the Washington Navy Yard (WNY), several Low Impact Development (LID) storm water management features have been installed throughout the region. LID features are designed to maintain and enhance, or recreate, the pre-development hydrologic regime of urban and developing watersheds. These LID features, such as bioretention cells and rain gardens, require specialized maintenance. It is expected that more LID features will be designed and constructed on regional facilities in the future, requiring incremental increases in the option years of this new contract.

OBJECTIVES

Due to the nature of LID design and materials, specialized care is needed to maintain their operational ability. The objective of this project is to establish a regular monitoring and maintenance program for the existing WNY LID features, with the flexibility to add the new features to the maintenance contract as they are constructed. Maintenance could include, but is not limited to: sediment removal; mulching; biomass removal; plant/tree replacement; structural repair; stone replacement; algae control; invasive species control; etc.

OVERALL CONDITIONS

As each option year is evaluated for renewal, NDW NAVFAC Washington Environmental will add any newly constructed LID features to the maintenance list. Additionally, it is the intention of NAVFAC Washington to modify the contract to expand LID maintenance to include all installations within the scope of this overall contract. The following types of LID features will be serviced:

Bioretention cells / rain gardens

Permeable Pavement (Pavers/Concrete/Asphalt)

Tree filters

Rain Barrels

Green Roofs

Ponds/Wetlands

Swales

PROJECT TASKS

NSA Washington

Task One: Establish LID Maintenance Requirements

NDW Environmental/NAVFAC Washington Environmental will provide the contractor with a list of LID features that will be included in the maintenance contract (attached spreadsheet). The contractor will visit all LID features with the NDW NSA Washington Storm Water Program Manager to ascertain their nature. Following these site visits, the contractor will submit the proposed maintenance regime for each existing LID feature to the NSA Washington Stormwater Program Manager for review. Example inspection checklists for various LID features can be found at <http://ddoe.dc.gov/swguidebook>.

Bioretention cells/Rain gardens:

- Inspect bioretention inlets quarterly for debris or clogging. Remove cover and clean debris as needed or annually.
- Remove weeds and mulch cells bi-annually. Scarify surface before re-mulching. Replace mulch to a depth of 2-3". Remove weeds and invasive materials. Fertilize in fall with slow release 5-10-5 fertilizer.
- Replace failed plantings when necessary with like kind-species and similar size.
- Remove collected sediments annually. Rake river rocks to the side to recover collected sediments.
- Replace river rocks as needed to ensure function as designed for energy dissipation.

Permeable pavers:

- Inspect pavers quarterly. Vacuum or remove any large debris or solids from openings. Remove cleanout for underdrains and inspect for sediments or clogging.
- NDW NAVFAC Washington will/may sweep openings yearly with street sweeper. Replace chip stones from openings. KTR shall provide all material for repair and replacement.

Street tree filter:

- Inspect inlet throat and grate quarterly. Remove debris as required. Remove grate and cleanout and dispose of collected sediment bi-annually. Remove trash, weeds, and debris as required. Fertilize plants in fall with slow release fertilizer 5-10-5 or equivalent.

Rain barrels:

- Inspect quarterly. Remove debris in down spout as required. Ensure proper functionality of barrel.

Task Two: LID Maintenance

The contractor will perform inspections and maintenance on all LID features within their contract, according to the regime approved by NDW Environmental. This will occur 1 time per quarter.

Task Three: Reporting

The contractor will, on a regular basis, email to the NDW NSA Washington a brief synopsis of what maintenance was performed on which date, and the current conditions of each LID feature. Through this process the contractor will alert NAVFAC Washington to any maintenance problems or performance issues with the LID features. This report will be provided quarterly, after the quarterly maintenance and inspection is accomplished.

POCs

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PLANTING ZONE 1

1. Planting zone 1 is located in the stream channel and along the banks and bed of this zone. Annual maintenance will be required to maintain herbaceous cover and prevent colonization by woody species, both of which are required to maintain drainage efficiency after construction.
 - a. At the end of the first growing season after construction, use a mower, string trimmer, or other mechanism to remove all woody species that grow along the bed and banks.
 - b. Throughout the growing season, conduct mowing and trimming within the entire zone to maintain vegetation to a height between 6 and 8 inches.
 - c. At the beginning of each growing season, woody species and invasive weeds are to be removed by hand or treated with an approved herbicide.
 - d. All mowing or trimming activities shall be conducted with care so as not to cause erosion or damage to infrastructure.

PLANTING ZONES 2 AND 4

1. Planting zone 2 is located along the top of banks with a purpose of replacing existing forest that was removed during the construction process and establishing riparian buffers in areas that will not interfere with existing utilities, infrastructure or operations. Planting zone 4 is similar to zone 2 but with an emphasis on lower growing trees and shrubs with more ornamental appeal.
 - a. Monthly during the first growing season mow the native grass / plant ground cover to remove invasive weed and woody species maintaining a height of 8 inches.
 - b. For the second year and beyond, mow the groundcover once annually at the beginning of the growing season.
 - c. At the beginning of the growing season each year, inspect for invasive weed or woody species and remove as needed in same manner as item C for zone one.
 - d. Care is to be taken not to damage any trees or shrubs planted intentionally.
2. During landscape installation, buck guards and other protective devices will be installed to protect trees and shrubs from deer damage.
 - a. These protective devices can be removed if they present a danger or begin to inhibit growth.

PLANTING ZONE 3

1. Planting zone 3 is a lawn area intended to provide easy maintenance access to infrastructure while protecting underground utilities from root damage and maintaining sight line near roads.

- a. This planting zone is to be mowed to a maximum of 6 inches throughout the growing season.

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Storm Water Management

Low Impact Development (LID) Maintenance Carderock

NSF Carderock

Task 01:

The contractor will visit the below listed LID features at Carderock with the NSA Washington Storm Water Program Manager to ascertain their nature. Following these site visits, the contractor will submit the proposed maintenance regime for each existing LID feature to the NSA Washington Stormwater Program Manager for review. Example maintenance requirements for various LID features can be found at in the Maryland Department of the Environment Stormwater Guidebook, available on-line at www.mde.state.md.us.

Green Roof of B42: 5,000 SF

Contractor will visit the site a minimum of seven (7) times through the spring and the fall. A maintenance visit shall entail:

1. Notifying government PAR of arrival and start of work for this purpose.
2. Green roof component inspection/ adjustment.
3. Manual watering where necessary.
4. Replace up to 10% of dead vegetation per year with new plants of similar kind bred for green roof plantings. Species shall be approved prior to application.
5. Inspection and clearing of all vegetation free zones.
6. Weed and invasive species removal.
7. Debris removal.
8. Up to two slow-release organic fertilizer applications, if needed
9. Drain inspection and clearing.
10. Documentation and submittal of maintenance services to Carderock PAR for each visit.
 - a. Documentation shall include but is not limited to:
 - i. Actions taken
 - ii. Concerns
 - iii. Suggestions to correct issues.

- iv. Type and quantity of materials used i.e. fertilizer.
- v. Number and types of plants replaced during visit.

North Parking Lot (1), B125 (3), Parade Ground (2) and Relocatable B (2) bioretention cells

- a. Remove all weeds monthly from April to November.
- b. Repair any eroded areas, quarterly.
- c. Check plant health monthly and identify plants replacement monthly. Replace in the spring or fall when planting conditions are appropriate.
- d. Prune plants annually, in spring or fall in accordance with species specific requirements.
- e. Maintain mulch at a depth of 3 inches. Scarify surface before remulching, annually
- f. Remove sediment from stone spillways annually, replace stone as needed per design
- g. Remove sediment and debris for outlets and outlet vaults, annually
- h. Ensure grass is growing around the parameter of the facility from mulch edge to 10 feet around the entire cell. Provide seed, stabilization, water and aeration , as need to ensure to ensure vegetative growth
- i. Inspect grass swales and conveyances annually for blockage sediment and erosion. Regrade, seed and stabilized as needed to ensure full grass coverage.
- j. Where rain garden seed mix is required by design, inspect for bare spots during the growing season and reseed, annually
- k. Report any broken or missing structural components or excessive deer browsing to Gov POC.

All materials to be provided by contractor.

B18 Sand Filter

- 1. Quarterly during growing season:
 - a. Remove all weeds from sand filter.
 - b. Rake sand bed throughout. Remove any foreign debris.
- 2. Annually:
 - a. Replace sand to a depth of 3 inches.
- 3. Annually: Remove manhole cover and without entering vault check depth of sediment on bottom of vault. Report depth of sediment to NSA Washington Stormwater Program Manager
- 4. All year long:

- a. Insure grass is growing at inlet (perimeter) to this facility from sand to 10 feet out around the entire filter and from bottom of forebays to 10 feet out. Provide seed, straw, fertilizer, aeration and whatever means necessary to ensure vegetative growth.
- b. Provide irrigation for establishing new growth when required.
- c. Repair damage due to erosion and stabilize soil and new growth efforts around all bays and dike walls.
- d. Inspect grass swales and conveyances annually for blockage, sediment and erosion. Regrade, seed and stabilize as needed to achieve full grass coverage.

Clean Curb Cuts.

Curb cuts at Building 19 round about, Building 18 parking lot and visitor's center parking lot. Clean curb cuts and concrete spillways of sediment, resetting and replacing stone as needed.

B198 Rain Barrels (4):

1. Quarterly:
 - a. Remove debris and dirt from inlet screen.
 - b. Ensure barrel is leveled and positioned directly over downspout.
 - c. Exercise drain valve at bottom of barrel and leave half-open.
 - d. When barrels become clogged for whatever reason, flush and clean as necessary and restore to proper functionality.

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General:

1. Discard debris collected from this maintenance off base.

J-1503050-13(f1)

GREEN ROOF MAINTENANCE – WRNMMC

Green Roofs on Bldgs. 17, 62, 64, and 82 at NSA Bethesda

Perform maintenance for the roof plants, roof beds and green roofs, maintenance to include the following:

1. Irrigate the beds and plants 1 to 1 1/2 inches to fully saturate the growing media. (In periods of drought, additional watering will be necessary).
2. Check that any permanent irrigation systems are functioning properly and make adjustments as needed.
3. Winterize the irrigation system if needed.
4. Make sure roof drains are functioning properly by clearing drains of any debris, including plant material soil and any other foreign objects.
5. Report any unsecured roof flashing and any waterproofing issues.
6. If green roof is overgrown with weeds at start of the period of performance, cut weeds as per COL 3 lawn cutting schedule to prevent overgrowth and root damage to roof.
7. Remove ALL plants from Vegetation Free Zones at perimeters, drains and other penetrations. All areas not designed for green roof growth must be kept free of vegetation to prevent root damage to roof system.
8. Inspect plant for overall health; look for signs of disease and pests.
9. Remove and properly dispose of any foreign objects, debris or garbage.
10. Remove any loose metal plates from GardMat Disk Anchors.

STORMWATER MAINTENANCE SCHEDULE BIORETENTION

Inspection Item	Frequency of Inspection	Inspection Requirements	Remedial Action
Bioretention Basin	Seasonally and after a major storm		
Dewatering		Facility must dewater within 48 hours of rainfall. Noticeable odors, stained water on the filter surface or at the outlet, or the presence of algae or aquatic vegetation are indicators of anaerobic conditions, and inadequate dewatering of the facility.	The top three inches of soil should be removed and replaced with soil material as per plan specifications. Follow up inspections must confirm adequate dewatering. If the facility does not function as intended after the above action, the entire filter and underdrain system may need maintenance. MOE approval may be necessary.
Mulch Layer		Check mulch for adequate cover, sediment accumulation, or discoloration.	Replace and remove old mulch and excess sediments. Provide adequate mulch cover according to approved design .
Vegetative Surfaces	Monthly		
Plant Composition and Health		Compare plant composition with approved plans. Check for invasive species or weeds. Check for dead or dying vegetation.	Remove and replace plants in accordance with plan specifications.
Vegetative Cover and Erosion		Check for evidence of erosion, runoff channelizing , or bare spots .	Re-seed or re-plant in accordance with approved landscaping plans. Re-grading may be required when concentrated flow causes rills or gullyng through

			the facility.
Debris and Trash Cleanout	Monthly	Check that the facility is clean of trash and debris. Inlets, outlets, and contributing areas around the facility must be checked .	Trash and debris must be disposed of in an acceptable manner according to current regulations .
Structural Components	Annually	Check for evidence of structural deterioration, spalling, or cracking. Inlet and outlet structures must be in good condition.	Repair to good condition according to specifications on the approved plans.

STORMWATER MAINTENANCE SCHEDULE BIORETENTION

Outlets	Seasonally and after a major storm	Check for evidence of erosion, rills, or gullyng.	Stabilize all eroded areas and grade to provide stable conveyance.
		Riprap outlet must be maintained in good functional condition.	Repair according to approved plan.
Pretreatment Forebays	Seasonally and after a major storm		
Sediment Accumulation		Check for sediment accumulation in the forebay.	When the forebay depth is less than half the proposed design, sediment must be removed and the forebay restored according to the approved design.
Sand Layer		Check sand for staining and sediment accumulation	Replace first three inches of sand layer with sand materials per plan specifications.

Gravel Diaphragm		Check gravel diaphragm for sediment accumulation and evidence of erosion	Stabilize or replace gravel according to plan specifications .
Grass Channel Conveyance Systems	Seasonally and after a major storm	Check for erosion, flow blockages, and stable conveyance	Stabilize and grade according to approved plan.
Overall Function of the Facility	Annually	Check that flow splitters are functioning as designed and that bypass is operating as designed.	Construction must be in accordance with approved plans.

* Field conditions may require a modification to the original approval in order to achieve the intended design function. Contact MDE's Sediment and Stormwater Management Plan Review Division at 410-537-3563 for review and approval of proposed modifications.